A Collaborative Network for Multiple Hyperspectral Images Joint Classification

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***Abstract*— In recent years, deep learning (DL) has achieved remarkable success in classifying hyperspectral images (HSIs), relying heavily on the quantity and quality of labeled samples. However, obtaining sufficient labels for HSIs poses a challenge. HSIs obtained by the same sensor often exhibit similar spectral information due to their shared physical, chemical properties, or reflective attributes. Joint analysis of several HSIs enables the integration of limited labeled samples and extraction of more robust and discriminative features from different HSIs. Therefore, a multitask collaborative network (MTCN) for the joint classification of multiple HSIs acquired by the same sensor in different areas is proposed. In the MTCN, each HSI has its own feature extraction channel, which facilitates the learning of image-specific representations. In addition, a feature sharing channel (FSC) is created to extract and transfer multihierarchical image-shared representations between multiple HSIs, thereby forming a common knowledge pool to facilitate feature sharing. Furthermore, a cross-channel mutual attention module (CMAM) is designed to collaboratively utilize features from image-specific and image-shared channels, enhancing the efficiency of information communication in HSIs. The experimental results on six HSIs demonstrate that the proposed MTCN can jointly classify multiple HSIs by the same sensor in different areas and achieve good classification performance.**

***Index Terms*— Attention mechanism, hyperspectral image (HSI) classification, joint analysis, multitask learning (MTL), multiple HSIs.**

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I. Introduction

HYPERSPECTRAL images (HSIs) have dozens or even hundreds of narrow and sequential spectral bands reflecting the material composition of ground objects [1]. HSI classification, which separates each class based on its rich spectral profiles, is widely studied and employed in many fields, including land resource management, environmental monitoring, military surveillance, and so on [2]. Many effective classification methods have already been put forth [3], [4]. However, certain unavoidable and unfavorable factors that seriously constrain classification accuracy exist in HSIs. These include the following: 1) high-dimensional spectral information, which contains a lot of useless data and interferes with the extraction of useful spectral features and 2) insufficient labels, which make it difficult to gather enough and trustworthy data to inform the classification process [5].

The utilization of reliable information concealed in HSI is enhanced by deep learning (DL), which learns to represent raw redundant data with informative feature representations. This approach [6], [7], [8], [9] has been proven effective in HSI classification. Specifically, convolutional neural networks (CNNs) have demonstrated excellent classification performance in remote sensing image processing due to their robust feature extraction and representation capabilities [10], [11], [12]. For instance, a CNN model was proposed to extract discriminative features from original data and generate classification results [13]. Gong et al. [14] proposed a multiscale CNNs (MS-CNNs) to extract multiscale features and deep metrics that maximize interclass variance while minimizing intraclass variance. However, the great performance of these methods is largely due to a large number of labeled samples. In the case of HSIs, acquiring sufficient labeled data is a challenging and time-consuming task that demands significant human effort and prior knowledge.

Facing the challenge of small training sets, several methods have constructed a discriminative feature space by designing sophisticated models, thereby facilitating hyperspectral classification based on limited labeled samples [15], [16], [17], [18], [19]. Sun et al. [15] proposed an unsupervised spectrum motion feature learning (SMF-UL) framework to learn spectral variation knowledge from unlabeled HSI samples for classification, and a small number of labeled samples were utilized to train the classifier. Braham et al. [16] introduced two positive pair-mining strategies within a contrastive learning framework to meet the demand for high-quality data augmentations.

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Sun et al. [17] proposed a spectral–spatial multiple layer perceptron (SSMLP)-like network to establish a fine-grained discriminative feature space, enhancing the classification performance of few-shot open-set HSI. Pal et al. [19] presented extreme value meta-learning to fit the Weibull distribution for each known class using a small number of support sets in order to achieve few-shot open-set hyperspectral classification. In scenarios characterized by limited labeled samples, these methods have devised innovative strategies to learn discriminative features that enable accurate hyperspectral classification.

In recent years, multitask learning (MTL), as a promising learning paradigm, has already been introduced into HSI classification to address the problem of insufficient labeled samples [20], [21], [22], [23], [24], [25]. In HSI classification, MTL establishes multiple relevant tasks and fully explores the intrinsic relationship between labeled and unlabeled samples. In MDL4OW [20], MTL was employed to simultaneously accomplish HSI classification and reconstruction tasks in an unfamiliar open environment, effectively preventing the overestimation of known land cover. Hang et al. [21] proposed a multitask generative adversarial network (MTGAN) for HSI classification, and the designed generator parallelly handled the reconstruction task and the classification task. Since the number of labeled and unlabeled samples was seriously unbalanced, Li et al. [23] designed a graph-based deep multitask few-shot learning (GDMFSL) for HSI classification to make the feature extraction of labeled and unlabeled samples converge at the same time. In these methods, the internal relationship of labeled and unlabeled samples was fully explored, thus improving the performance of HSI classification. However, the methods established multiple tasks for a single HSI containing limited information.

At present, a large number of cross-scene hyperspectral classification algorithms have been extensively studied. These algorithms aim to utilize labeled samples from the source HSI to guide the classification of the target HSI, which has limited labeled samples [26], [27], [28], [29]. Huang et al. [26] proposed an adversarial domain adaptation network to align the source and target feature spaces through weighted adversarial discrimination. Due to the challenge of acquiring labels, Li et al. [27] proposed a dynamic adaptation network to extract spatial–spectral joint features from both source and target HSIs. Zhang et al. [28] proposed a language-aware domain generalization network that employed a two-stream architecture to extract visual and linguistic features. These methods effectively learn domain-invariant representations from the source HSI and enhance the classification performance of the target HSI. However, these methods ignore the classification performance of the HSI in the source domain.

For HSI classification, the spectral information has its own physical meaning (radiance or reflectance) and does not depend on the HSI used. According to hyperspectral imaging, different HSIs obtained by the same sensor have analogous spectral information, and the joint analysis of multiple HSIs acquired by the same sensor in different areas has a positive effect on extracting potential common features, enabling to promote the representation ability of each HSI classification task. Furthermore, the issue of scarce labeled samples can be alleviated by sharing labeled data with diverse HSI classification tasks.

Therefore, a multitask collaborative network (MTCN) model with specific and shared channels is proposed to cooperatively address several HSI classification. The contributions of this model can be summarized as follows.

1) An MTCN model is proposed for the joint classification of multiple HSI datasets acquired by the same sensor, enabling adaptive collaborative analysis and shared knowledge extraction across various HSI classification tasks. This approach addresses the challenge of limited labeled samples in certain hyperspectral datasets by leveraging annotations from diverse HSI datasets.
2) To comprehensively explore potential common knowledge, the feature sharing channel (FSC) and the cross-channel mutual attention module (CMAM) are specially designed. Depart from image-specific features learned by separate HSI channel, FSC focuses on extracting significant image-shared features. In addition, CMAM promotes effective communication between separate HSI channel and FSC by selectively transferring features with attentional correlation between them.
3) In the experiments, the better performance than compared methods verify the MTCN enables to collaboratively classifying multiple HSIs at the same time and promotes the classification performance of each HSI task based on its common representation extraction and sharing capabilities. The classification performance rises with the growing number of tasks verify the superiority of multiple HSIs joint classification.

The remainder of this article is organized as follows. Section II presents the background and motivation of this study. Section III describes the proposed method in detail. Section IV presents the experiment results, and the conclusion is drawn in Section V.

II. Background and Motivation

This section demonstrates the basic framework of HSI classification, which facilitates to directly understand studying content. In addition, the basic paradigm of the proposed method, multitask learning (MTL), has been described in detail. Finally, we summary the motivations of the proposed method to highlight the rationality and innovation.

A. Traditional DL-Based HSI Classification

The diagram of traditional DL-based HSI classification framework is shown in Fig. 1 and is described in detail as follows.

1) Data Description: As shown in Fig. 1(a), in the HSI data $\mathcal{I}$, the 3-D HSI data cubes, composed of center pixels and their neighbor pixels, are grouped by the same batch size before being fed into the network. The label of each cube corresponds to the label of the central pixel.

2) Feature Extraction: In Fig. 1(b), all 3-D cubes with real labels are selected to train the neural network and learn the significant features of raw data. After training, discriminative representations can be learned by deep neural network.

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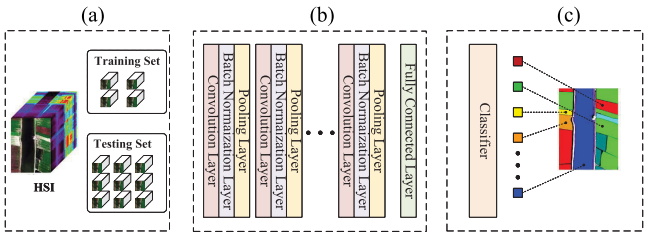


Fig. 1. Traditional CNN-based HSI classification. (a) Data description. (b) Feature extraction. (c) Classification result.

3) Classification Result: For HSI classification, the outputs of the network are the pixel labels. The deep representations are input in full connection layer to get the classification probability, and different land-cover categories are assigned to cubes.

It is important to note that extracting highly separable features from a set of limited number of labeled samples remains difficult in the current classification models based on deep networks.

B. Multitask Learning

MTL has gained significant attention from researchers in various domains due to its numerous advantages. One of the key benefits of MTL is its ability to leverage shared representations and knowledge across multiple related tasks [30]. Given m tasks $T_1, T_2, \ldots, T_m$, each task independently trains a different network in single-task learning (STL). However, in MTL, the training data from all m tasks are used to train a single learning model. By jointly analyzing multiple tasks, the model can capture common patterns and dependencies, leading to improved generalization and performance [31]. In addition, MTL allows for efficient data utilization, as it can leverage the abundance of labeled data from one task to improve the performance on other related tasks with limited labeled data [32]. This not only reduces the need for large labeled datasets but also alleviates the burden of labeling. Moreover, MTL promotes knowledge sharing, where information learned from one task can be transferred and applied to improve performance on other tasks. This flexibility and adaptability make MTL a promising approach in various domains, including computer vision [33], natural language processing [34], and bioinformatics and health informatics [35], where it has achieved remarkable results in solving complex classification problems.

C. Motivation of the Proposed Method

1) Joint Collaboration of Multiple HSIs: In the case of insufficient labeled samples, most existing methods [36], [37], [38] achieve classifying pixels by use sample augmentation or delicate models. Zhu et al. [37] designed the H-B sampling strategy that is used to generate random stratified training sample data, which solve the problem of insufficient and unbalanced samples. Feng et al. [38] proposed a geometric regularized Laplacian low-rank representation-based graph, which reveals the global low-rank and local spatial structure and achieves accurately classifying mixed pixels. In reality, as shown in Fig. 2, according to hyperspectral

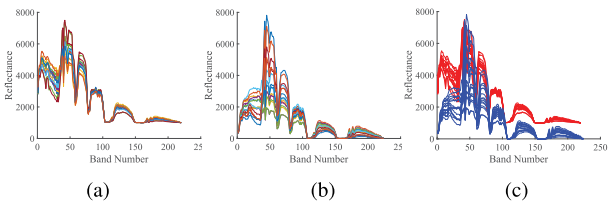


Fig. 2. Reflectance of two HSIs (Indian Pines and Salinas) obtained by the AVIRIS sensor. (a) IP. (b) SA. (c) IP (red) and SA (blue).

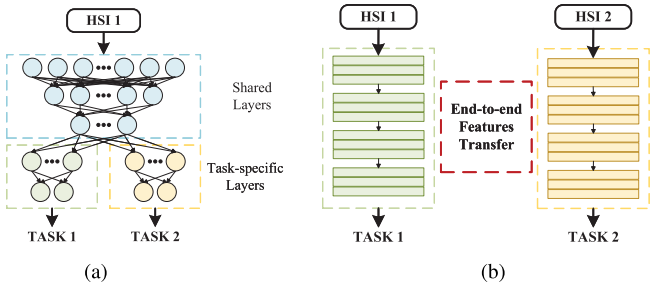


Fig. 3. Motivation of collaborative network. (a) Traditional MTL model. (b) MTL model for multiple HSIs.

imaging mechanism, HSIs obtained by the same sensor have a similar spectral range [39]. Most existing methods ignore the intrinsic relationship among these HSIs from the same sensor, resulting in learning of limited useful features from a single HSI. Therefore, the joint analysis of multiple HSIs acquired by the same sensor in different areas is accessible and necessary.

2) Motivation of Collaborative Network: As shown in Fig. 3(a), most models [40], [41] analyzing one HSI at a time only shared common knowledge in the shallow part (encoder/generator) and separately analyzed labeled and unlabeled samples in other parts of the network. Xue et al. [40] proposed an MTL architecture to learn multimodal robust features, in which the weights of feature representation networks for each multimodal data were shared. A hybrid MTL network [41] was proposed for HSI classification, where a backbone network was designed to handle an HSI classification task and a spatial mask classification task simultaneously. In addition, traditional models [42], [43] analyzed multiple HSIs by a supernet and extracted potential shared knowledge by hard-parameter sharing. Liu and Shi [43] proposed a multitask DL method for HSI classification. The method trained an identical feature extractor for all the HSI data. However, these methods ignored the differences in correlation between different datasets from distinct areas, which could negatively impact the classification accuracy of each HSI. Therefore, as shown in Fig. 3(b), it is worth designing a model for multiple HSIs that automatically adjusts the ability to learn common knowledge based on the correlation between HSIs.

III. Methodology

Most HSIs lack labeled samples, making it difficult to train a well-performing classification network for each HSI. Therefore, an MTCN with specific and shared channels is proposed to simultaneously address multiple HSI classification tasks from different areas and fully exploit image-specific and image-shared features in different channels, respectively. Moreover, MTCN enables exchange useful information in

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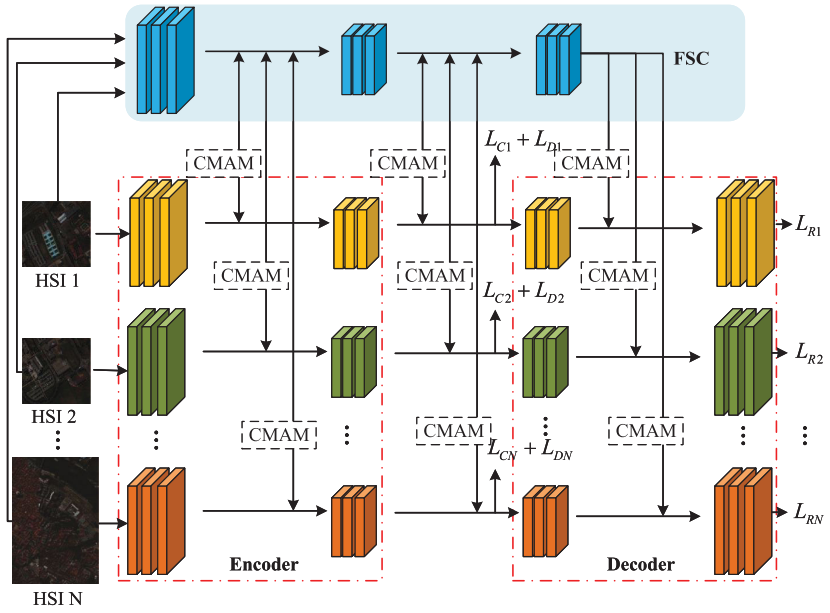


Fig. 4. Flowchart of MTCN.

Algorithm 1 Training of MTCN (There Are N HSI Datasets $\{\mathcal{I}_1, \ldots, \mathcal{I}_N\}$, Each Containing a Different Number of Labeled Samples)

Input: Unlabeled data $\boldsymbol{X}_1^u, \ldots, \boldsymbol{X}_N^u$; Labeled data $\boldsymbol{X}_1^l, \ldots, \boldsymbol{X}_N^l$;
Output: Trained MTCN;
1: Initialize the parameters of network;
2: **while** termination condition is not satisfied **do**
3: **For each HSI classification tasks $\mathcal{I}_n$:**
4: a) Unlabeled data $\boldsymbol{X}_n^u$ are input into MTCN and obtain reconstruction data $\hat{\boldsymbol{X}}_n^u$;
5: b) Labeled data $\boldsymbol{X}_n^l$ are fed into encoder and obtain features z_n';
6: c) Calculate loss L_n;
7: Optimization of loss $L = \sum_{n=1}^{N} L_n$.
8: **end while**

different HSIs from the same sensor, thereby enhancing the classification accuracy of each HSI. The flowchart of MTCN is shown in Fig. 4, and the pseudocode is listed in Algorithm 1.

A. Whole Framework of the Proposed Method

Assuming multiple HSIs $\{\mathcal{I}_1, \ldots, \mathcal{I}_N\}$, $\mathcal{I}_n \in \mathbb{R}^{W_n \times H_n \times B}$ acquired by the same sensor in different areas are obtained. $\mathcal{I}_n \in \mathbb{R}^{W_n \times H_n \times B}$, W_n, H_n, and B denote the length, width, and number of bands of $\mathcal{I}_n$. Each HSI has a different number of labels. $\boldsymbol{X}_n^u$ and $\boldsymbol{X}_n^l$ denote the unlabeled and labeled data from $\mathcal{I}_n$. In the proposed MTCN, each HSI $\mathcal{I}_n$ is established as independent classification tasks T_n. As shown in Fig. 4, each HSI classification task T_n has its own image-specific feature extraction channel. The hyperspectral samples corresponding to each task T_n $(n = 1, \ldots, N)$ are fed into these channels separately. The upper image-shared feature extraction channel, named FSC, is used to obtain potential common knowledge. It takes as input the concatenation of the inputs from the image-specific feature extraction channels.

These feature extraction channels have the same network structure, and each channel is a simple AE, which enables capture internal structural information by reconstructing unlabeled data. Each channel is combined with an encoder and decoder. The encoder is composed of four convolution layers, and the decoder is formed by three deconvolution layers. The specific configuration of the different layers of one of the network channels in MTCN is shown in Table I. In the MTCN, the transfer of significant features does not occur directly between the image-specific feature extraction channels, but instead through the FSC and CMAM.

It is worth noting that the structure of the FSC is not exactly the same as that of the image-specific channels. Compared with the image-specific channels, the FSC does not have the last layer of the AE structure. The FSC merely serves as a place for the exchange of common knowledge among the tasks and does not have any independent output. The common feature extraction ability of FSC is determined by all the tasks together.

In the MTCN, distinct feature spaces are generated by the encoders. The image-specific feature extraction channels generate the image-specific feature spaces $\boldsymbol{Z}_n$ containing the high-level representations of $\mathcal{I}_n$. The FSC generates the image-shared feature space $\boldsymbol{Z}_s$, which reflects the potential common knowledge among all the HSIs $\{\mathcal{I}_1, \ldots, \mathcal{I}_N\}$. Based on these feature spaces, classifiers assign land-cover class for each pixels. Besides, image-specific decoders generate reconstruction HSIs $\{\hat{\mathcal{I}}_1, \ldots, \hat{\mathcal{I}}_N\}$.

B. Feature Sharing Channel

Since multiple HSIs acquired by the same sensor in different areas have image-shared and image-specific representations, learning both features simultaneously poses a challenge.

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TABLE I
SPECIFIC STRUCTURE OF A CHANNEL IN MTCN

	Layers	Input shape	Filter size	Activation function	BN	Stride	Padding	Output shape
Encoder	Conv1	$(1, S, S, B)$	$(32, S, S, 3)$	LeakyReLU	Y	1	Y	$(32, 1, 1, B)$
	Conv2	$(32, 1, 1, B)$	$(32, 1, 1, 3)$	LeakyReLU	Y	2	Y	$(32, 1, 1, B/2)$
	Conv3	$(32, 1, 1, B/2)$	$(32, 1, 1, 3)$	LeakyReLU	Y	2	Y	$(32, 1, 1, B/4)$
	Conv4	$(32, 1, 1, B/4)$	$(32, 1, 1, 3)$	LeakyReLU	Y	2	Y	$(32, 1, 1, B/8)$
Decoder	Deconv1	$(32, 1, 1, B/8)$	$(32, 1, 1, 4)$	LeakyReLU	Y	2	Y	$(32, 1, 1, B/4)$
	Deconv2	$(32, 1, 1, B/4)$	$(32, 1, 1, 4)$	LeakyReLU	Y	2	Y	$(32, 1, 1, B/2)$
	Deconv3	$(32, 1, 1, B/2)$	$(1, S, S, 4)$	Tanh	Y	2	Y	$(1, S, S, B)$

Therefore, in this article, an FSC is proposed to effectively learn image-shared features, which is established as task T_s.

The FSC extracts multihierarchical common features of the original data. The input cubes to the FSC are a concatenation of the inputs from image-specific channels. Furthermore, the FSC provides image-shared information for image-specific tasks and obtains distinct image-specific representations from these tasks. It is worth noting that the transmission of features does not occur only at one stage, but throughout the entire process of feature extraction.

Specifically, it is assumed that after feature extraction at layer i, the output of each image-specific feature extraction channels is denoted as f_n^i $(n = 1, \ldots, N)$, and the output of the FSC is denoted as f_s^i. Through the proposed attention module, the features transferred from the image-specific channel to the FSC are represented as $\hat{f}_n^i$, and the features transmitted from the FSC to each image-specific channel are represented as $\hat{f}_s^i$. Then, the features extracted by the next convolution layer $i + 1$ are expressed as follows:

$$\begin{aligned} f_n^{i+1} &= \mathcal{F}_n^{i+1}\left(f_n^i \oplus \hat{f}_s^i\right) \\ f_s^{i+1} &= \mathcal{F}_s^{i+1}\left(f_s^i \oplus \hat{f}_1^i \oplus \cdots \oplus \hat{f}_N^i\right) \end{aligned} \tag{1}$$

where f_n^{i+1} is the output of image-specific channels at layer $i + 1$. f_s^{i+1} is the output of the FSC at layer $i + 1$. $\oplus$ denotes elementwise concatenation. $\mathcal{F}_n^{i+1}(\cdot)$ and $\mathcal{F}_s^{i+1}(\cdot)$ represent convolution operation. Through the above cross-channel information interaction, more supplementary features $\hat{f}_n^i$ are fed into the layer $i + 1$ of FSC, thereby learning more complete common representations. For image-specific channels, significant common features $\hat{f}_s^i$ are fed into layer $i + 1$, contributing to learning representative features. Taking two HSIs as an example, Fig. 5 is a schematic of the cross-channel communication of features.

The FSC extracts and combines features from image-shared and image-specific channels as it progressively learns potential common features. This allows for multihierarchical representations interaction and enhances in-depth communication between tasks $\{T_1, \ldots, T_N\}$.

C. Cross-Channel Mutual Attention Module

In the proposed model, image-specific feature extraction channels do not deliver information directly, but transfer image-shared knowledge through the FSC. Therefore, abundant features are fed into the FSC, resulting in significant representations easily get lost in the abundance of redundant features. The attention mechanism is able to determine

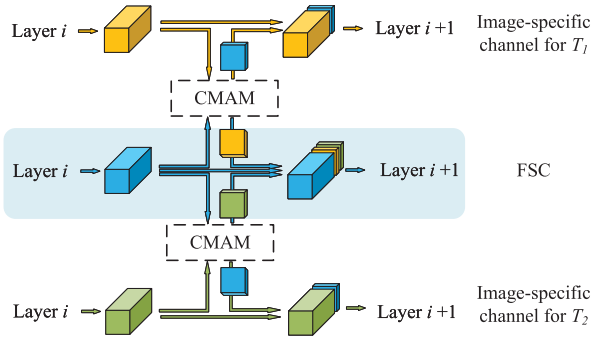


Fig. 5. Cross-channel communication of features.

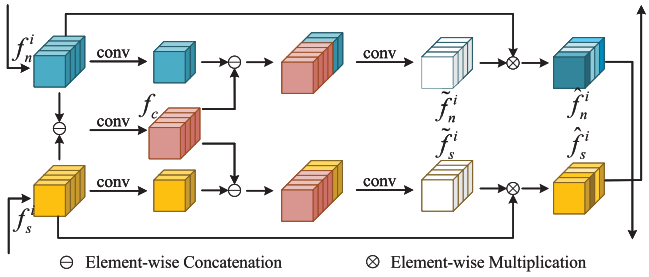


Fig. 6. Cross-channel mutual attention module.

which parts of the features to prioritize and allocate limited data processing resources to important features. Traditional attention modules are employed within channels rather than across channels, and they only use their own information to determine which part of the features to focus on. Therefore, the traditional attention modules are not suitable for cross-task communication. Taking the above into account, as shown in Fig. 6, the CMAM is designed. It is added to the communication process between the FSC and image-specific channels.

Specifically, suppose that after completing a convolution operation of layer i, the output features of image-specific channels and FSC are expressed as f_n^i, $f_s^i \in \mathbb{R}^{c\times 1\times 1\times b}$, which are as the input of CMAM. The output of the CMAM $\hat{f}_n^i$, $\hat{f}_s^i \in \mathbb{R}^{c\times 1\times 1\times b}$ can be formulated as follows:

$$\hat{f}_n^i, \hat{f}_s^i = \mathcal{F}_{\text{CMAM}}\left(f_n^i; f_s^i; \theta_a\right) \tag{2}$$

where $\mathcal{F}_{\text{CMAM}}(\cdot)$ denotes the CMAM and θ_a represents its parameters. The CMAM consists of three steps: obtaining fusion features, generating relation maps, and producing mutual features. The module details are as follows.

1) Obtaining Fusion Features: Due to the differences in input data, it is crucial to mine the underlying common

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knowledge between the data. Therefore, first, the fusion features f_c of the input data are obtained through elementwise concatenation and convolution operations

$$f_c = \mathcal{F}\left(f_n^i \oplus f_s^i, \theta_c\right) \tag{3}$$

where θ_c is a collection of parameters of $\mathcal{F}(\cdot)$.

2) Generating Relation Maps: Fusion features f_c contain both image-specific and image-shared information, with an emphasis on the potential common knowledge between the two input data. Furthermore, in order to further explore the different mutually beneficial relationships between the input data, f_c is elementwise concatenated with the processed input data. Moreover, the relation maps are generated by convolution operation. The generation process of relation maps $\widetilde{f_n^i}$ and $\widetilde{f_s^i}$ is as follows:

$$\begin{aligned}\widetilde{f_n^i} &= \mathcal{F}\left(f_c \oplus \mathcal{F}\left(f_n^i, \theta_n\right), \theta_{mn}\right)\\ \widetilde{f_s^i} &= \mathcal{F}\left(f_c \oplus \mathcal{F}\left(f_s^i, \theta_s\right), \theta_{ms}\right).\end{aligned} \tag{4}$$

3) Producing Mutual Features: The features $\hat{f}_n^i$ and $\hat{f}_s^i$ are acquired through the elementwise multiplication of the input data f_n^i, f_s^i and the relation maps $\widetilde{f_n^i}$, $\widetilde{f_s^i}$. This can be expressed as follows:

$$\begin{aligned}\hat{f}_n^i &= f_n^i \otimes \widetilde{f_n^i}\\ \hat{f}_s^i &= f_s^i \otimes \widetilde{f_s^i}\end{aligned} \tag{5}$$

where $\otimes$ represents elementwise multiplication. $\hat{f}_n^i$ $(n = 1, \ldots, N)$ containing the information required by the FSC is provided to the FSC. $\hat{f}_s^i$ including the information beneficial to the image-specific tasks is provided to the image-specific channels.

The CMAM is in action whenever image-specific channels communicate with the FSC. The CMAMs enable fully explore the correlation of features from image-shared and image-specific channels. According to the different mutually beneficial relationships, CMAMs achieve adaptively selecting cross-channel informative feature representations for image-shared and image-specific channels.

D. Loss of MTCN

In order to make the MTCN fully learn the image-specific and image-shared features, the loss function is designed from the following two aspects: 1) MTCN needs abundant samples to learn high-level image-specific and image-shared representations and 2) MTCN enables extract sufficient discriminative features with limited labeled samples. Therefore, based on reconstruction loss, classification loss, and distance metric, the loss function L of the proposed MTCN can be expressed as follows:

$$\begin{aligned}L_n &= L_{Rn} + L_{Cn} + L_{Dn}, \quad n = 1, \ldots, N\\ L &= \sum_{n=1}^{N} L_n\end{aligned} \tag{6}$$

where L_n represents the loss of the task T_n. L_{Cn} represents the classification loss of the task T_n. L_{Dn} represents the distance metric of the task T_n. L_{Rn} represents the reconstruction loss of the task T_n.

Considering a huge amount of parameters to be trained, how to mobilize more available resource to stimulate the great modeling ability of a complex network is essential. Therefore, the proposed model takes advantages of ample unlabeled samples in HSIs to learn high-level representations. For task T_n, we add the reconstruction loss L_{Rn} and express it as follows:

$$L_{Rn} = \frac{1}{M_n}\sum_{j=1}^{M_n}(x_{nj} - \hat{x}_{nj})^2 \tag{7}$$

where $\hat{x}_{nj} \in \hat{\boldsymbol{\mathcal{I}}}_n$ represents the reconstructed samples and $x_{nj} \in \boldsymbol{\mathcal{I}}_n$ denotes the raw samples. M_n denotes the number of pixels.

In the MTCN, image-specific and image-shared features play a distinct role in collaboratively classifying multiple HSIs acquired by the same sensor in different areas. Therefore, the features used to assign a label for each pixel are the concatenation of image-shared and image-specific features, which are as follows:

$$\boldsymbol{z}_n' = \boldsymbol{z}_s \oplus \boldsymbol{z}_n \tag{8}$$

where $\boldsymbol{z}_n$ represents image-specific features from $\boldsymbol{Z}_n$ and $\boldsymbol{z}_s$ denotes image-shared features from $\boldsymbol{Z}_s$. Based on the $\boldsymbol{z}_n'$, the predicted probability p_{nj}^c that the observed pixel j belongs to class c can be formulated as follows:

$$\begin{aligned}p_{nj}^c &= \mathrm{FC}(z_{nj})\\ z_{nj} &\in \boldsymbol{z}_n'\end{aligned} \tag{9}$$

where $\mathrm{FC}(\cdot)$ denotes the fully connected layers. Therefore, the expression for the classification loss L_{Cn} is

$$L_{Cn} = -\frac{1}{M_n}\sum_{j=1}^{M_n}\sum_{c=1}^{C} y_{nj}^c \log\left(p_{nj}^c\right) \tag{10}$$

where C denotes the number of categories. y_{nj}^c is a symbolic function that takes a value of 0 or 1. If the true class of the sample j is equal to c, it takes a value of 1; otherwise, it takes a value of 0.

It is challenging to understand and summarize the link between features and labels due to a lack of labeled pixels. By calculating the distance between samples, metric learning is skilled at quantifying intraclass compactness and interclass separability, which enhances classification performance with a small number of labeled pixels. Therefore, the metric loss L_{Dn} can be expressed as follows:

$$L_{Dn} = \frac{1}{M_n}\sum_{j=1}^{M_n}\max\left(D\left(z_{nj}'^a, z_{nj}'^+\right) - D\left(z_{nj}'^a, zc_{nj}'^-\right) + \mathrm{margin}, 0\right) \tag{11}$$

where $z_{nj}'^a$ denotes the anchor features and $z_{nj}'^+$ represents positive features, namely, features with the same label as the anchor features. Also, $z_{nj}'^-$ is negative features, namely, features with a different label with the anchor features. $D(\cdot,\cdot)$ is the Euclidean distance. margin is the minimum gap between $D(\boldsymbol{z}_n^a, \boldsymbol{z}_n^+)$ and $D(\boldsymbol{z}_n^a, \boldsymbol{z}_n^-)$. $\max(\cdot,\cdot)$ means take the maximum value.

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Based on the loss L, the model takes full advantage of more accessible and helpful data, and the MTCN can fully mine potential common knowledge between multiple HSIs from the same sensor and extract discriminant features conducive to classification.

IV. Experiment

In this section, the experimental datasets are described. Moreover, we present the experimental setup for training and evaluating the proposed model. Finally, comparative results with other methods, parameter analysis, ablation experiments, and scalability experiment are conducted.

A. Experimental Datasets

In the experiments, six public HSI datasets are employed, including the Pavia University (PaviaU), Pavia Center (PaviaC), WHU-Hi-LongKou (LongKou), WHU-Hi-HongHu (HongHu), Dioni, and Loukia datasets. When simultaneously classifying different HSIs acquired by the same sensor in different areas, PaviaU and PaviaC datasets from the same sensor collaboratively are undertaken. LongKou and HongHu datasets obtained by the same sensor are jointly classified. Dioni and Loukia datasets from the same sensor are collaboratively classified.

1) PaviaU and PaviaC Datasets: PaviaU and PaviaC were photographed by the ROSIS satellite sensor. PaviaU was located at the University in Pavia, Italy. Also, PaviaC was taken over Pavia, northern Italy. PaviaU is a 610 × 340 pixel image, and PaviaC is a 1096 × 715 pixel image. The 103 bands were used to complete the classification in PaviaU. The PaviaC dataset has 102 spectral bands. The PaviaU and PaviaC contain nine land-cover categories. Fig. 7(a) and (e) shows the false-color image and the corresponding ground truth. The false-color image and ground truth of PaviaC are shown in Fig. 7(b) and (f).

2) LongKou and HongHu Datasets: The LongKou and HongHu were obtained by Headwall Nano-Hyperspec imaging sensor in Longkou Town and Honghu City, respectively, in Hubei Province, China. They have 270 bands. LongKou and HongHu images are 550 × 400 and 940 × 475 pixels in size, respectively. LongKou and HongHu contain 9 and 22 land-cover classes, respectively. The false-color image and ground truth of LongKou are shown in Fig. 7(c) and (g). HongHu's false-color image and ground truth are shown in Fig. 7(d) and (h).

3) Dioni and Loukia Datasets: The datasets were collected by Hyperion sensor (EO-1, USGS). They have 176 bands. The Dioni and Loukia are 250 × 1376 and 249 × 945 pixels in size, respectively. Their area includes 12 different land-cover categories. Dioni's false-color image and ground truth are shown in Fig. 7(i) and (k). Loukia's false-color image and ground truth are shown in Fig. 7(j) and (l).

B. Experimental Settings

To verify the efficiency of the proposed method, eight algorithms are used for comparison. The comparison methods are mainly divided into nonjoint and joint classification methods.

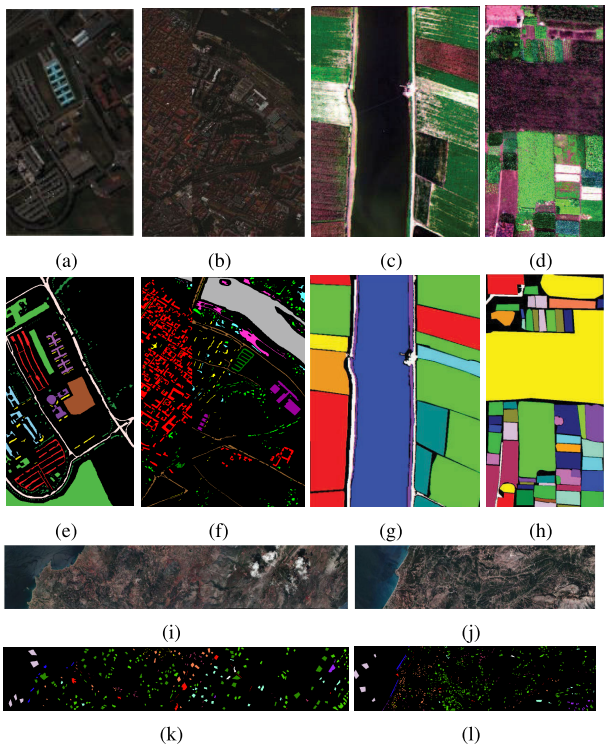


Fig. 7. Six HSI datasets. (a) and (e) PaviaU. (b) and (f) PaviaC. (c) and (g) LongKou. (d) and (h) HongHu. (i) and (k) Dioni. (j) and (l) Loukia.

The nonjoint methods are further divided into the supervised methods (HybridSN [8] and ResNet [9]), the semisupervised methods (HSGAN [44] and 3DAES [45]), and few-shot methods (DCFSL [46] and DM-MRN [47]). The joint classification methods contain MMoE [48] and MTAN [49]. Each compared method used in our experiments is configured in the following way.

1) HybridSN: Three "Conv3d" convolution layers, one "Conv2d" convolution layer, and a final linear layer are included. Also, the training epochs, learning rate, and batch size are set to 100, 0.001, and 256, respectively.

2) ResNet: The main architecture of the network is the same as that of HybridSN. The attention-based adaptive spectral–spatial kernel module is added between "Conv3d" convolutions. Also, the training epochs, learning rate, and batch size are set to 100, 0.001, and 256, respectively.

3) HSGAN: It consists of a generator and a discriminator. The generator contains two fully connected layers and two 1-D convolutional layers. The discriminator contains 1-D convolutional layers and one fully connected layer. Also, the training epochs, learning rate, and batch size are set to 200, 0.0005, and 128, respectively.

4) 3DAES: It contains four "Conv3d" convolution layers and four "Deconv3d" deconvolution layers. Also, the training epochs, learning rate, and batch size are set to 200, 0.0001, and 128, respectively.

5) DCFSL: The DCFSL contains two "3-D residual blocks" to extract source and target features, and a conditional domain-adaptive feature module including five fully connected layers is adopted to achieve domain alignment. Also, the training epochs and learning rate are set to 1000 and 0.001, respectively.

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TABLE II
CLASSIFICATION ACCURACY (%) FOR THE PAVIAU IMAGE (AVERAGE OF TEN RUNS)

Class	200 labeled samples per class					20 labeled samples per class						
	Non-joint methods		Joint methods			Non-joint methods				Joint methods		
	HybridSN	ResNet	MMoE	MTAN	MTCN	HSGAN	3DAES	DCFSL	DM-MRN	MMoE	MTAN	MTCN
Asphalt	96.35	92.56	83.09	98.33	**98.69**	84.53	81.58	90.46	93.02	82.82	82.11	**88.09**
Meadows	99.49	99.20	91.06	99.52	**99.92**	91.30	87.06	92.19	97.52	66.76	84.43	**97.74**
Gravel	77.98	88.66	84.23	94.09	**97.86**	86.77	89.32	**91.82**	80.09	72.37	86.42	90.00
Trees	95.85	98.17	97.00	99.02	**99.18**	91.95	**95.63**	94.81	91.55	75.75	94.35	95.23
Metal	99.70	99.84	99.93	**100.00**	**100.00**	**100.00**	99.92	99.72	**100.00**	99.33	**100.00**	**100.00**
Bare Soil	97.55	99.24	90.73	99.82	**100.00**	90.20	96.45	97.72	**99.09**	73.57	90.55	97.10
Bitumen	99.39	99.19	92.41	99.70	**99.92**	98.09	99.24	93.21	90.89	78.95	97.37	**99.92**
Bricks	84.35	96.00	89.30	96.50	**97.45**	83.07	67.64	72.86	**91.80**	75.50	78.71	87.75
Shadows	74.23	94.38	97.78	**99.89**	**99.89**	95.04	93.74	99.24	96.73	85.11	**98.52**	97.89
OA	95.60 ±0.78	97.22 ±0.57	90.19 ±1.24	98.84 ±0.96	**99.37** ±0.83	89.81 ±2.53	87.30 ±2.17	91.50 ±4.03	94.59 ±3.29	73.53 ±3.59	86.31 ±3.44	**94.89** ±3.27
AA	91.66 ±0.73	96.36 ±0.49	91.73 ±1.76	98.54 ±0.88	**99.21** ±1.23	91.22 ±2.96	90.07 ±1.69	92.45 ±3.95	93.06 ±3.02	78.91 ±4.59	90.27 ±4.56	**94.86** ±4.24
Kappa	94.16 ±0.84	96.21 ±0.62	87.18 ±2.04	98.46 ±1.26	**99.17** ±1.74	86.64 ±3.68	87.29 ±2.32	88.87 ±4.19	**94.57** ±4.17	80.45 ±5.06	82.34 ±5.13	93.25 ±4.91

TABLE III
CLASSIFICATION ACCURACY (%) FOR THE PAVIAC IMAGE (AVERAGE OF TEN RUNS)

Class	200 labeled samples per class					20 labeled samples per class						
	Non-joint methods		Joint methods			Non-joint methods				Joint methods		
	HybridSN	ResNet	MMoE	MTAN	MTCN	HSGAN	3DAES	DCFSL	DM-MRN	MMoE	MTAN	MTCN
Water	98.79	**100.00**	99.51	**100.00**	**100.00**	96.72	99.39	99.88	**100.00**	99.15	99.76	99.88
Trees	97.07	96.46	96.10	97.44	**99.39**	92.80	96.22	92.44	85.49	71.22	94.76	**95.37**
Asphalt	**99.63**	96.81	94.61	99.51	99.51	92.40	**95.10**	93.14	93.14	62.38	94.12	**95.10**
Bricks	99.88	99.50	99.50	**100.00**	**100.00**	**99.63**	97.15	99.50	97.52	46.91	98.76	97.40
Bitumen	97.40	98.02	95.67	99.50	**100.00**	91.21	97.03	93.44	97.40	90.59	92.45	**98.39**
Tiles	98.97	97.38	96.90	99.52	**99.68**	81.59	90.24	96.27	98.10	76.98	**98.81**	94.29
Shadows	**99.79**	99.16	88.45	99.37	99.58	94.12	90.34	92.86	**96.64**	58.40	91.60	93.91
Meadows	98.42	**99.88**	98.67	93.81	98.67	94.66	96.12	**98.91**	97.94	84.59	97.45	97.94
Bare Soil	96.46	98.54	97.32	**98.78**	98.54	90.12	97.56	96.71	**99.39**	95.49	91.95	94.39
OA	98.46 ±0.69	98.32 ±0.55	96.70 ±0.94	98.67 ±0.84	**99.49** ±0.89	91.86 ±2.76	95.39 ±2.08	96.07 ±1.53	96.27 ±1.61	77.11 ±3.46	95.90 ±2.99	**96.29** ±3.05
AA	98.49 ±0.74	98.42 ±0.49	96.30 ±1.13	98.66 ±0.79	**99.49** ±0.93	92.58 ±2.71	95.46 ±2.07	95.91 ±2.07	96.18 ±2.11	76.19 ±3.87	95.52 ±1.80	**96.30** ±2.36
Kappa	98.00 ±0.81	99.17 ±0.60	97.10 ±1.42	99.12 ±1.12	**99.31** ±1.25	92.24 ±2.79	96.55 ±2.22	97.43 ±1.95	**97.87** 2.24	76.23 ±3.97	96.89 ±3.17	97.36 ±3.51

6) DM-MRN: The DM-MRN is composed by an embedding module (four 2-D convolutional layers), an image-to-class metric module (three 2-D convolutional layers), and an image-to-image metric module (four 2-D convolutional layers). Also, the training epochs and learning rate are set to 1000 and 0.005.

7) MMoE: The MMoE contains four "Conv3d" convolution layers, three "Deconv3d" deconvolution layers, and one fully connected layer. Also, the training epochs, learning rate, and batch size are set to 600, 0.001, and 200, respectively.

8) MTAN: It contains four "Conv3d" convolution layers, three "Deconv3d" deconvolution layers, and one fully connected layer. Also, the training epochs, learning rate, and batch size are set to 600, 0.001, and 200, respectively.

In MTCN, the network is optimized via Adam optimizer, the training epochs are 600, the learning rate was set to 0.001, the batch size was 200, and the EPS was $1e^{-08}$. The nonjoint methods are carried out on six datasets, respectively. For joint classification methods, PaviaU and PaviaC datasets are simultaneously involved in classification. LongKou and HongHu are another pair of HSIs used for classification. Dioni and Loukia datasets are simultaneously involved in classification. When PaviaU (PaviaC) has 200 labeled samples per class, PaviaC (PaviaU) has 20 labeled samples per class. When the number of labeled samples is 200 per class on LongKou (HongHu), the number of labeled samples is 20 per class on HongHu (LongKou). When Dioni (Loukia) has 200 labeled samples per class, Loukia (Dioni) has 20 labeled samples per class.

All experiments were conducted on a server equipped with four Intel[1] Xeon[1] Silver 4214R CPU @ 2.40 GHz, 192 GB of DDR4 RAM, and eight NVIDIA Tesla V100s GPU. Regarding our software environment, it is composed by Ubuntu 5.4.0-6Ubuntu1 16.04.10 as an operating system, CUDA 10.2 and cuDNN 7.6.5, and PyTorch framework and with Python 3.9.13 as the programming language.

To quantify the effectiveness of the proposed method, three indicators are adopted to measure the classification results. The indicators include overall accuracy (OA), average accuracy (AA), and Kappa coefficients (Kappa).

C. Experimental Results

Tables II–VII show the quantitative assessment of six HSI datasets, respectively, and the classification maps are shown in Figs. 8–13.

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TABLE IV
CLASSIFICATION ACCURACY (%) FOR THE LONGKOU IMAGE (AVERAGE OF TEN RUNS)

Class	200 labeled samples per class					20 labeled samples per class						
	Non-joint methods		Joint methods			Non-joint methods				Joint methods		
	HybridSN	ResNet	MMoE	MTAN	MTCN	HSGAN	3DAES	DCFSL	DM-MRN	MMoE	MTAN	MTCN
Corn	99.47	**99.94**	98.32	98.09	99.23	96.81	73.17	96.45	96.98	90.61	97.45	**97.91**
Cotton	93.72	95.22	85.75	97.43	**97.60**	93.64	99.84	95.03	**99.10**	87.46	93.67	90.52
Seasame	98.32	86.34	95.48	98.94	**99.93**	92.11	99.77	96.34	98.85	98.85	99.08	**99.74**
B.Soybean	**97.02**	95.77	88.70	91.56	96.01	86.94	76.70	**93.34**	89.79	75.13	86.65	87.82
N.Soybean	72.27	94.24	92.39	98.31	**99.52**	97.23	98.17	93.76	**99.21**	92.27	98.77	99.11
Rice	99.03	97.93	99.63	**99.87**	99.08	**99.50**	81.49	87.86	90.45	99.57	99.32	99.29
Water	99.65	**99.99**	99.74	99.94	99.76	97.16	83.16	**99.80**	98.35	**99.80**	99.04	99.79
Roads	90.54	**99.56**	92.18	97.96	98.03	91.63	62.39	70.59	84.26	90.69	**94.33**	93.77
Weed	82.44	83.82	93.78	**96.88**	96.54	**96.12**	43.37	54.50	86.82	88.95	93.15	93.50
OA	97.20 ±1.34	97.62 ±1.29	94.88 ±2.58	96.74 ±3.01	**98.24** ±4.13	93.64 ±4.52	78.87 ±4.34	94.00 ±3.17	94.29 ±5.06	89.35 ±5.06	94.42 ±5.19	**94.98** ±5.79
AA	92.50 ±1.45	94.76 ±1.67	94.00 ±3.16	97.66 ±2.81	**98.41** ±3.75	94.57 ±4.79	79.78 ±5.61	87.52 ±4.55	93.76 ±5.94	91.48 ±5.94	**95.72** ±6.37	**95.72** ±6.27
Kappa	95.94 ±1.39	96.94 ±2.04	93.36 ±4.42	95.75 ±4.79	**97.70** ±3.99	91.76 ±5.01	72.45 ±6.73	93.47 ±3.61	93.26 ±6.77	86.36 ±6.77	92.81 ±6.61	**93.49** ±7.81

TABLE V
CLASSIFICATION ACCURACY (%) FOR THE HONGHU IMAGE (AVERAGE OF TEN RUNS)

Class	200 labeled samples per class					20 labeled samples per class						
	Non-joint methods		Joint methods			Non-joint methods				Joint methods		
	HybridSN	ResNet	MMoE	MTAN	MTCN	HSGAN	3DAES	DCFSL	DM-MRN	MMoE	MTAN	MTCN
Roof	**98.08**	97.68	94.05	96.22	**98.08**	64.14	**95.56**	89.14	84.00	50.97	67.89	86.08
Road Soil	87.73	74.94	86.67	97.89	**98.18**	42.40	70.76	77.59	**96.55**	61.13	64.41	71.16
Bare	93.19	**93.98**	77.64	87.81	88.74	70.98	81.01	77.19	83.47	84.86	68.29	**86.48**
Cotton	89.50	**99.60**	89.36	94.09	96.28	**93.11**	92.19	88.20	90.79	46.87	45.96	88.91
CottonW.	**99.31**	51.06	78.22	95.82	96.62	32.92	92.18	84.72	87.25	59.34	60.00	**92.91**
Rape	93.31	**95.46**	89.34	94.32	95.03	85.39	80.11	90.12	**91.67**	53.11	59.65	91.35
C.Cabbage	**93.20**	87.23	62.82	72.66	83.54	48.79	**60.96**	55.60	57.08	32.36	33.51	57.86
Pakchoi	94.35	55.40	43.49	93.46	96.55	62.28	64.33	45.76	46.28	32.91	37.49	**81.03**
Cabbage	96.03	97.18	94.93	96.59	**97.68**	90.61	93.86	93.33	**95.79**	74.95	89.45	89.27
Mustard	89.03	89.46	64.92	85.36	**91.68**	55.34	78.33	**80.21**	68.74	69.20	67.63	78.18
B.chinensis	86.86	80.64	59.11	85.87	**94.35**	22.41	39.02	**74.85**	60.52	30.69	34.69	72.16
Brassica	79.19	61.47	63.71	73.75	**82.43**	56.86	70.99	**75.39**	66.14	64.99	38.90	60.29
S.Brassica	71.61	84.89	63.06	76.68	**86.17**	42.76	47.54	58.85	**62.23**	48.97	42.34	65.04
Lactuca	98.19	89.61	80.17	92.14	**93.75**	59.48	85.06	84.23	**85.36**	51.81	53.87	74.32
Celtuce	99.00	86.93	90.52	98.70	**99.30**	92.91	95.81	90.52	**97.90**	49.60	89.62	94.51
FC.Lettuce	**99.57**	93.69	84.03	93.58	94.38	71.59	88.76	90.68	**91.23**	51.16	65.12	89.15
R.Lettuce	98.60	93.82	81.76	**99.27**	**99.27**	72.49	96.01	84.92	96.21	65.48	66.98	**96.54**
Carrot	97.79	90.95	88.37	**98.91**	98.35	67.73	94.87	84.40	93.94	61.77	65.09	**99.35**
Radish	95.31	92.56	85.96	92.92	**95.71**	91.04	90.58	82.69	**93.48**	38.69	53.97	92.70
Garlic	97.22	75.27	89.96	**97.30**	97.05	43.55	88.78	87.49	92.71	37.61	35.94	**93.75**
B.Bean	99.47	32.08	77.18	**99.77**	**99.77**	44.43	99.10	89.76	85.77	35.02	62.35	**99.17**
Tree	98.04	79.93	84.98	97.28	**98.09**	37.03	94.88	95.67	94.41	59.80	57.77	**96.76**
OA	90.54 ±2.94	92.56 ±3.47	82.42 ±4.28	90.73 ±4.13	**93.92** ±3.89	75.68 ±5.68	82.70 ±5.73	82.58 ±4.74	84.09 ±4.29	51.02 ±6.80	51.55 ±6.33	**84.11** ±7.28
AA	93.39 ±2.31	81.99 ±3.45	78.65 ±5.61	91.84 ±4.92	**94.59** ±4.30	61.28 ±6.16	81.85 ±5.23	80.97 ±4.61	82.80 ±4.03	52.79 ±6.99	57.31 ±7.14	**84.41** ±8.46
Kappa	88.32 ±2.47	90.51 ±3.38	78.31 ±6.12	88.45 ±5.03	**92.39** ±5.09	61.15 ±7.05	72.34 ±5.46	80.12 ±4.38	79.95 ±3.96	49.57 ±7.48	50.36 ±6.73	**80.41** ±9.45

TABLE VI
CLASSIFICATION ACCURACY (%) FOR THE DIONI IMAGE (AVERAGE OF TEN RUNS)

Class	200 labeled samples per class					20 labeled samples per class						
	Non-joint methods		Joint methods			Non-joint methods				Joint methods		
	HybridSN	ResNet	MMoE	MTAN	MTCN	HSGAN	3DAES	DCFSL	DM-MRN	MMoE	MTAN	MTCN
OA	94.93 ±1.38	**97.30** ±0.89	86.22 ±1.76	79.61 ±2.07	95.41 ±1.06	80.58 ±3.92	82.15 ±3.61	61.82 ±2.30	82.49 ±2.06	74.13 ±4.59	82.43 ±4.38	**88.72** ±2.28
AA	89.32 ±1.09	93.40 ±0.81	87.86 ±2.19	83.37 ±1.89	**96.33** ±0.94	82.46 ±4.24	86.07 ±3.98	73.83 ±2.49	89.39 ±2.14	73.72 ±5.81	84.17 ±4.82	**91.92** ±2.65
Kappa	93.36 ±1.43	96.24 ±1.04	85.99 ±3.01	80.52 ±2.65	**96.36** ±1.38	81.39 ±4.51	83.45 ±4.17	58.90 ±1.97	81.05 ±2.02	72.66 ±4.94	84.50 ±4.95	**86.06** ±1.99

1) PaviaU Dataset: The quantitative classification accuracy results are listed in Table II, and the classification maps of compared methods on the PaviaU dataset are visually shown in Fig. 8.

With 200 labeled samples per class, the proposed method has the best classification performance, achieving an OA of 99.37%. In comparison with MMoE and MTAN, other methods yield superior classification results. It indicates that

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TABLE VII
CLASSIFICATION ACCURACY (%) FOR THE LOUKIA IMAGE (AVERAGE OF TEN RUNS)

Class	200 labeled samples per class					20 labeled samples per class						
	Non-joint methods		Joint methods			Non-joint methods				Joint methods		
	HybridSN	ResNet	MMoE	MTAN	MTCN	HSGAN	3DAES	DCFSL	DM-MRN	MMoE	MTAN	MTCN
OA	87.43 ±1.57	90.53 ±1.14	87.56 ±2.48	87.06 ±2.25	**90.90** ±1.86	71.67 ±3.76	74.34 ±3.26	84.51 ±2.71	86.16 ±2.38	70.75 ±4.16	71.72 ±3.74	**88.76** ±3.02
AA	86.44 ±2.03	90.46 ±1.72	91.28 ±3.11	90.52 ±1.89	**95.95** ±2.01	75.04 ±3.95	79.83 ±2.97	89.32 ±3.19	90.29 ±2.76	75.97 ±3.88	79.36 ±3.14	**90.62** ±2.86
Kappa	85.82 ±1.91	**91.37** ±2.10	90.62 ±2.73	89.43 ±2.46	89.05 ±1.40	74.69 ±3.77	72.16 ±3.51	78.16 ±2.40	81.30 ±2.80	73.81 ±4.29	80.09 ±4.20	**89.72** ±3.11

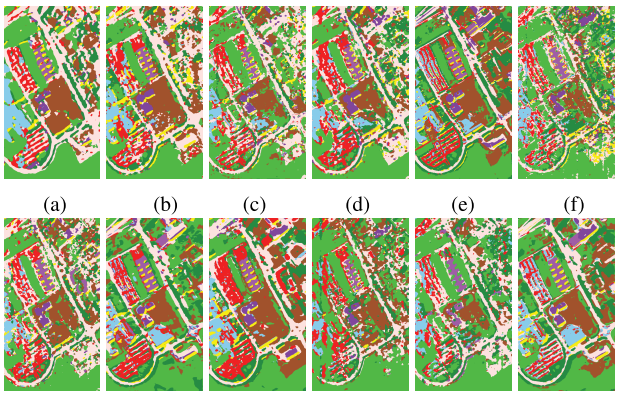


Fig. 8. Classification maps on the PaviaU dataset. (a) HybridSN. (b) ResNet. (c) MMoE(200). (d) MTAN(200). (e) MTCN(200). (f) HSGAN. (g) 3DAES. (h) DCFSL. (i) DM-MRN. (j) MMoE(20). (k) MTAN(20). (l) MTCN(20).

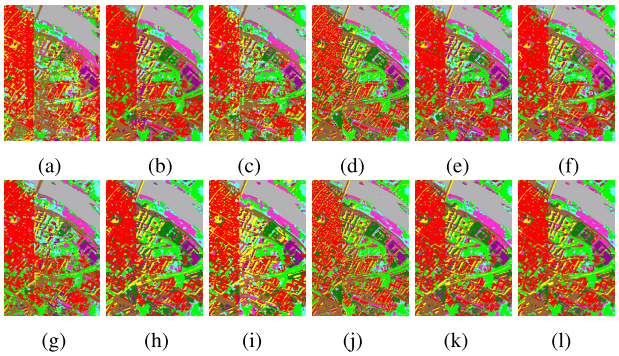


Fig. 9. Classification maps on the PaviaC dataset. (a) HybridSN. (b) ResNet. (c) MMoE(200). (d) MTAN(200). (e) MTCN(200). (f) HSGAN. (g) 3DAES. (h) DCFSL. (i) DM-MRN. (j) MMoE(20). (k) MTAN(20). (l) MTCN(20).

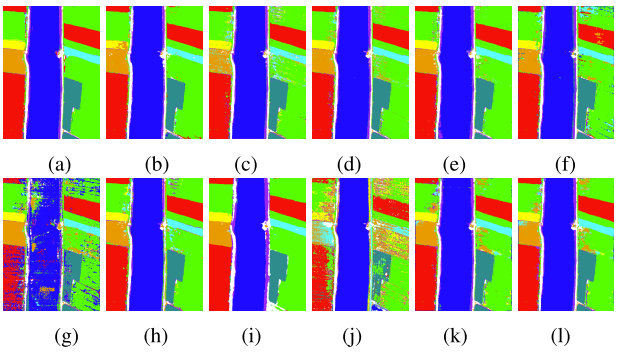


Fig. 10. Classification maps on the LongKou dataset. (a) HybridSN. (b) ResNet. (c) MMoE(200). (d) MTAN(200). (e) MTCN(200). (f) HSGAN. (g) 3DAES. (h) DCFSL. (i) DM-MRN. (j) MMoE(20). (k) MTAN(20). (l) MTCN(20).

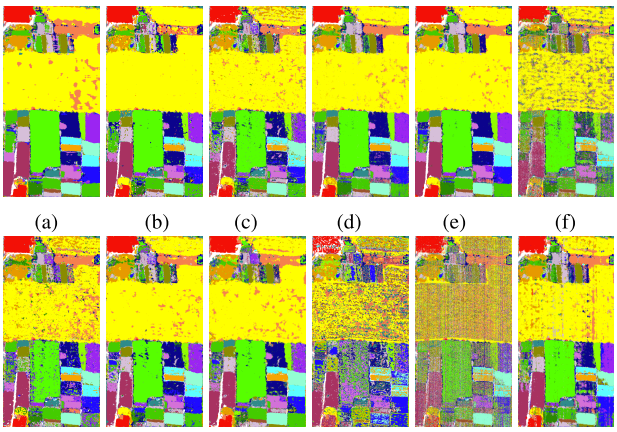


Fig. 11. Classification maps on the HongHu dataset. (a) HybridSN. (b) ResNet. (c) MMoE(200). (d) MTAN(200). (e) MTCN(200). (f) HSGAN. (g) 3DAES. (h) DCFSL. (i) DM-MRN. (j) MMoE(20). (k) MTAN(20). (l) MTCN(20).

MMoE and MTAN cannot effectively manage the relationships among multiple HSIs, resulting in negative knowledge transfer. Furthermore, the proposed method outperforms HybridSN and ResNet, showcasing its capability to extract shared knowledge and enhance feature representation.

When the number of labeled samples is 20 per class, the proposed method demonstrates the best classification performance, with an OA of 94.89%. In comparison with MMoE and MTAN, other methods show superior classification performance. This demonstrates that MMoE and MTAN cannot extract useful common features among HSIs. The proposed method outperforms semi-supervised methods (HSGAN and 3DAES) and few-shot methods (DCFSL and DM-MRN) in terms of classification performance. The comparative experimental results highlight that the proposed method effectively addresses multiple HSI classification tasks, thereby improving the classification performance across all tasks.

2) PaviaC Dataset: Table III and Fig. 9 show the quantitative and visual classification results of different methods on the PaviaC dataset.

In Fig. 9(a)–(e), it is evident that MMoE incorrectly classifies the Water class as Asphalt class and Shadows class, respectively. According to Table III (supervised classification), HybridSN, ResNet, and MTAN have similar classification performance. MTCN has the highest classification accuracy in

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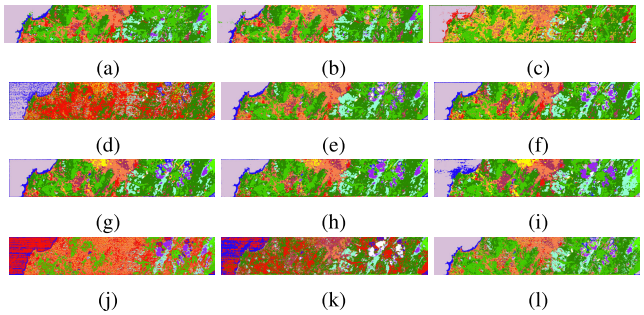


Fig. 12. Classification maps on the Dioni dataset. (a) HybridSN. (b) ResNet. (c) MMoE(200). (d) MTAN(200). (e) MTCN(200). (f) HSGAN. (g) 3DAES. (h) DCFSL. (i) DM-MRN. (j) MMoE(20). (k) MTAN(20). (l) MTCN(20).

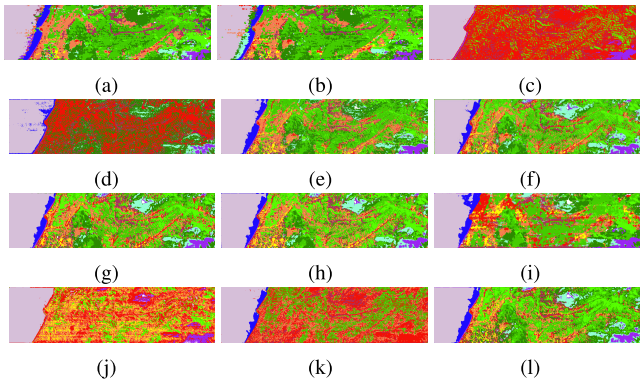


Fig. 13. Classification maps on the Loukia dataset. (a) HybridSN. (b) ResNet. (c) MMoE(200). (d) MTAN200. (e) MTCN(200). (f) HSGAN. (g) 3DAES. (h) DCFSL. (i) DM-MRN. (j) MMoE(20). (k) MTAN(20). (l) MTCN(20).

five classes, and its OA, AA, and Kappa are the best, reaching 99.52%, 99.49%, and 99.31%.

From the view of Fig. 9(f)–(l) and Table III (20 labeled samples per class), 3DAES and MTAN have better accuracy index and classification map in comparison with MMoE and HSGAN. The classification performance of DCFSL, DM-MRN, and MTCN is similar. DM-MRN has the highest Kappa (97.87%), and MTCN has the highest OA (96.29%) and AA (96.30%). Overall, the proposed MTCN has great classification performance on the PaviaC dataset.

3) LongKou Dataset: The visual results of the LongKou dataset are displayed in Fig. 10, and the quantization results are shown in Table IV.

As shown in Fig. 10(a)–(e) and Table IV (200 labeled samples per class), MTCN has the best quantitative indicators, and its OA, AA, and Kappa values are 98.24%, 98.41%, and 98.41%, respectively. MMoE has obvious misclassification for the Cotton class. HybridSN, ResNet, and MTAN have the highest classification accuracy in some classes. For example, MTAN achieves the best classification performance in Rice and Weed classes.

As shown in Fig. 10(f)–(k) and Table IV (20 labeled samples per class), it is easily found that HSGAN, DCFSL, and DM-MRN are obviously better than 3DAES, whose OAs are 93.64%, 94.00%, and 94.29% in nonjoint methods. Compared with MMoE, MTCN and MTAN can better deal with classification task on the LongKou dataset and obtain higher classification accuracy.

4) HongHu Dataset: Fig. 11 reports the classification maps of different methods on the HongHu dataset, and Table V displays its quantitative indicators.

It can be seen from Fig. 11(a)–(e) and Table V (200 labeled samples per class) that ResNet has a great performance in the Cotton class. HybridSN and MMoE mistakenly classify numerous pixels belonging to the Cotton class as the Cotton W. class. The classification accuracy of MTAN is similar to HybridSN. Overall, MTCN has the best classification performance, and its OA, AA, and Kappa are 93.92%, 94.59%, and 92.39%.

In Fig. 11(f)–(j) and Table V (20 labeled samples per class), it is obvious that there exist many misclassification pixels in all classification maps. The proposed MTCN has the best classification performance with an OA of 84.11% among all the methods. In nonjoint methods, although their OA and AA are not the best, they have the best classification accuracy in some classes, such as the Cotton and Celtuce classes. For joint methods MMoE and MTAN, their classification performance is worse than nonjoint methods. This is because they ignore the complex relationship of different datasets from distinct areas, which leads to the negative transfer of information.

5) Dioni Dataset: The visual results of the Dioni dataset are displayed in Fig. 12, and the quantization results are shown in Table VI.

With 200 labeled samples per class, other methods demonstrate better classification performance compared with MMoE and MTAN. This demonstrates that MMoE and MTAN cannot learn the shared representation effectively by utilizing the same backbone network. Furthermore, ResNet outperforms HybridSN in classification accuracy, indicating that the intricate structure of ResNet aids in learning valuable discriminative features. ResNet achieves the highest OA at 97.30%. The proposed algorithm attains the highest AA at 96.33%, highlighting its ability to leverage labeled samples from other datasets to enhance the classification accuracy of challenging classes.

When there are 20 labeled samples per class, the proposed method shows the highest classification performance, with an OA value of 88.72%. In comparison with MMoE, MTAN selectively learns multihierarchical representations from backbone networks using the attention mechanism, enhancing feature representation capabilities. In contrast to DCFSL, DM-MRN utilizes deep local descriptors to further explore the deep relationships among features. The semi-supervised algorithms HSGAN and 3DAES effectively leverage unlabeled samples, achieving the OAs of 80.58% and 82.15%. The comparative analysis reveals that the proposed method optimally utilizes multiple datasets, acquiring specific and shared representations, thereby enhancing the classification performance on the Dioni dataset.

6) Loukia Dataset: The visual results of the Loukia dataset are displayed in Fig. 13, and the quantization results are shown in Table VII.

When there are 200 labeled samples per class, MMoE, MTAN, and HybridSN exhibit similar classification performance. ResNet displays good classification performance due

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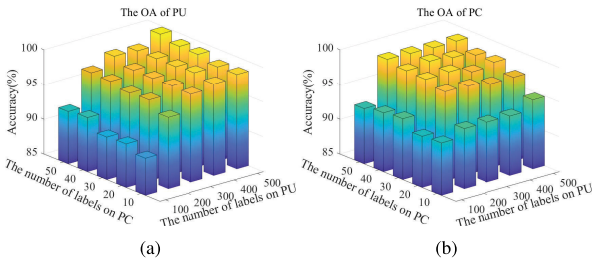


Fig. 14. Analysis of the influence of training samples per class on classification accuracy. (a) PaviaU. (b) PaviaC.

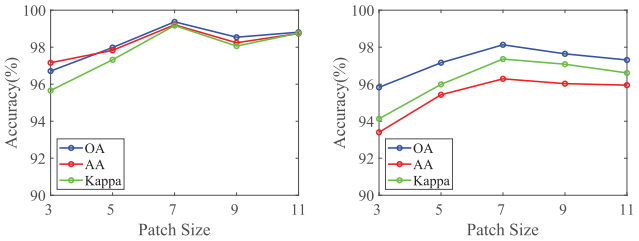


Fig. 15. Analysis of the influence of patch size. (a) PaviaU. (b) PaviaC.

to its intricate feature extraction network. The proposed algorithm, on the other hand, achieves the highest OA of 90.90% by leveraging valuable common knowledge from other datasets.

When labeled samples are 20 per class, HSGAN, 3DAES, MMoE, and MTAN exhibit similar classification performance. In contrast, for the few-shot algorithms DCFSL and DM-MRN, their ability to construct a feature space that accurately captures the relationships between different samples results in higher classification accuracy. The proposed algorithm not only captures relationships across multiple datasets but also establishes a common knowledge space, enhancing the model's ability to represent different samples. As a result, it achieves the highest classification accuracy of 88.76%.

D. Analysis of the Parameters

To examine the impact of the number of training samples and the spatial size of input cubes on HSI classification performance, the PaviaU and PaviaC are used as examples in this part.

1) Number of Training Samples: In HSI classification, the number of labeled samples directly affects the training and representation ability of the model. Therefore, it is necessary to evaluate the impact of different labeled samples on the classification performance of the proposed model MTCN. The labeled samples of PaviaU are set from the range of (100, 200, 300, 400, 500) per class, and the labeled samples of the PaviaC are set from the range of (10, 20, 30, 40, 50) per class.

The relationship between the OA values of PaviaU and PaviaC and the change in the number of training samples in MTCN are shown in Fig. 14. When the number of labeled samples for PaviaU increases from 100 to 200 per class, the OA values of PaviaU and PaviaC are greatly improved, meaning MTCN learns more significant features that facilitate PaviaU and PaviaC classification tasks. When the number of labeled samples for PaviaU is greater than 200 per class, the growth rate of OA slows down. It expresses that 200 labeled samples per class are the best number of labels for PaviaU to balance the classification performance of MTCN and the difficulty of the labeled samples. When the number of labels is 200 per class for PaviaU, with the increase of labeled samples for PaviaC, the inflection point is 20 per class. It can be found that the number of labels ranges from 10 to 20, and PaviaC has a greater increase in OA than PaviaU. Therefore, 200 per class for the PaviaU and 20 per class for the PaviaC are chosen for MTCN.

2) Spatial Size of Input Cubes: The spatial size of the input cubes is an important factor affecting the classification results. Insufficient window space can result in limited classification performance due to the lack of adequate spatial information. A window with excessive spatial data will introduce an abundance of diverse label data, thereby interfering with accurate classification. The spatial size of input cubes is set from the range of (3×3, 5×5, 7×7, 9×9, 11×11).

When PaviaU has 200 labeled samples per class and PaviaC has 20 labeled samples per class, Fig. 15 illustrates how the quantitative indicators of PaviaU and PaviaC follow the change of the spatial size of input cubes in MTCN. When the spatial size of input cubes is 7×7, the quantitative indicators on PaviaU and PaviaC are the best, reaching about 99% and 98%, respectively. The results show that the classification accuracy of PaviaU and PaviaC improves as the spatial size of input cubes increases from 3×3 to 7×7. However, the OA values of PaviaU and PaviaC slightly decrease with the spatial size increase from 7×7 to 11×11. It demonstrates that small spatial size introduces insufficient useful spatial information, which leads to the limitation of classification performance. In addition, large spatial size brings lots of significant spatial information, but interference spatial information is brought, which makes the classification accuracy decrease. In the experiments, 7×7 is selected as the spatial size of input cubes in MTCN.

E. Ablation Experiment Analysis

The proposed model achieves multiple HSI joint classification (MHJC) by feature sharing. The framework has one task-shared channel, namely, FSC, and multiple task-specific channels. FSC learns common features from different HSIs. In addition, the CMAM selectively transfers features between FSC and every image-specific feature extraction channel. The setting of extensive ablation experiments by adding different components is shown in Table VIII. The explanation of all abbreviations in Table VIII is as follows.

1) Single HSI Classification (SHC): The SHC uses an image-specific classification networks to complete single hyperspectral classification task.

2) Multiple HSIs Joint Classification (MHJC): In MHJC, these image-specific networks directly share features with each other, and features are fused by elementwise concatenation.

3) Feature Sharing Channel: FSC is designed to explore the common knowledge across multiple hyperspectral classification tasks. All image-specific classification networks share features with FSC.

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TABLE VIII
ANALYSIS OF THE EFFECTIVENESS OF DESIGNED STRATEGIES ON PAVIAU AND PAVIAC

Methods	Components			Accuracy(%)	
	MHJC	FSC	CMAM	PU	PC
SHC	×	×	×	94.48	97.54
MHJC	✓	×	×	87.09	93.87
MHJC+FSC	✓	✓	×	94.85	97.56
MHJC+CMAM	✓	×	✓	91.96	96.83
MTCN(Ours)	✓	✓	✓	**99.37**	**98.14**

TABLE IX
VERIFICATION EXPERIMENTS OF SCALABILITY

Datasets	Methods	OA	AA	Kappa
PaviaC	Single dataset	98.67	97.13	97.32
	Two datasets	99.28	98.15	97.86
	Three datasets	**99.63**	**98.72**	**98.87**
PaviaU1	Single dataset	90.13	91.81	90.66
	Two datasets	93.50	92.39	92.24
	Three datasets	**94.37**	**93.12**	**93.93**
PaviaU2	Single dataset	91.29	92.13	91.82
	Two datasets	93.34	93.71	93.65
	Three datasets	**93.97**	**94.23**	**94.15**

4) Cross-Channel Mutual Attention Module: The CMAM enables to selectively share features between FSC and image-specific networks according to the correlation between cross-task features.

As shown in Table VIII, it can be seen that the performance of the proposed model greatly improves compared with that of SHC; the classification accuracy increases by about 5% on the PaviaU dataset, and the proposed model has the highest classification accuracy 98.14% on the PaviaC dataset. This shows that the proposed model can transfer and share information among multiple tasks to improve the robustness of the model. In addition, the classification performance of MHJC and MHJC + CMAM is inferior to SHC, which means negative transfer easily happens in multiple HSIs from different areas joint classification. Furthermore, MHJC + FSC and proposed model outperform that SHC, and it indicates that the designed strategies can effectively use the relevant data of multiple tasks to learn the underlying shared representation and transfer useful common knowledge to image-specific channels.

F. Verification of Scalability

We have conducted comparative experiments on single-dataset classification, two-dataset collaborative classification, and three-dataset collaborative classification on the PaviaC, PaviaU1, and PaviaU2 datasets. The PaviaC dataset is classified with 200 labeled samples per class, and PaviaU1 and PaviaU2 are classified with 20 labeled samples per class.

The classification results are shown in Table IX. It can be seen that the three-dataset classification has the best classification performance, whose OAs of PaviaC, PaviaU1, and PaviaU2 are 99.63%, 94.37%, and 93.97%, respectively. It indicates that the proposed method can collaboratively address more than two HSIs and can increase the available data and improve the classification performance on each dataset by processing more datasets collaboratively.

V. CONCLUSION

In this article, the MTCN for the joint classification of multiple HSIs acquired by the same sensor in different areas is introduced. Image-specific and image-shared features are extracted using different channels in the triple-channel model. In the MTCN, the FSC learns and shares multihierarchical common features, which facilitates the extraction of potential relationships among different HSIs. In addition, CMAM is integrated into the information exchange process, effectively adaptively selecting significant representations to transfer among different channels.

Despite the classification performance is improved by the MTCN, it is worth noting that FSC and CMAM, which are designed for common knowledge extraction and sharing, have complex structures. As a result, a significant amount of computational resources is required. In addition, this article only focuses on the same classification tasks being performed on different datasets. It cannot consider the scenario where different datasets have distinct tasks, such as classification and segmentation tasks. Therefore, designing a lightweight and task-agnostic network model suitable for joint analysis of multiple HSIs from different areas poses an issue worth studying.

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